

數值方法_作業六_E14101082_陳政謙

1. Use Gaussian elimination and pivoting technique to solve

$$1.19x_1 + 2.11x_2 - 100x_3 + x_4 = 1.12$$

$$14.2x_1 - 0.112x_2 + 12.2x_3 - x_4 = 3.44$$

$$100x_2 - 99.9x_3 + x_4 = 2.15$$

$$15.3x_1 + 0.110x_2 - 13.1x_3 - x_4 = 4.16$$

```
PS C:\Users\user\Desktop\E14101082_numerical_hw6> cd "c:\Users\user\Desktop\E
Q1 Gaussian solution:
x1 = 0.176776
x2 = 0.0126921
x3 = -0.0206612
x4 = -1.18326
```

2. Find the inverse of the matrix A where

$$A = \begin{bmatrix} 4 & 1 & -1 & 0 \\ 1 & 3 & -1 & 0 \\ -1 & -1 & 6 & 2 \\ 0 & 0 & 2 & 5 \end{bmatrix}$$

```
PS C:\Users\user\Desktop\E14101082_numerical_hw6> cd "
0.279693 -0.0804598 0.0383142 -0.0153257
-0.0804598 0.37931 0.0574713 -0.0229885
0.0383142 0.0574713 0.210728 -0.0842912
-0.0153257 -0.0229885 -0.0842912 0.233716
```

3. Use Crout factorization for a tri-diagonal system to solve the problem

$$\begin{bmatrix} 3 & -1 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}.$$

```
PS C:\Users\user\Desktop\E14101082_numerical_hw6> cd "c
x1 = 1.43636
x2 = 2.30909
x3 = 2.49091
x4 = 1.16364
```